AmFall: WiFi CSI Amplitude-Based Fall Detection Using Denoised Scalograms

Yongkeun Kim, *Member, IEEE*, Wha Sook Jeon, *Senior Member, IEEE*, and Dong Geun Jeong, *Life Senior Member, IEEE*

***Abstract*—Radio frequency (RF)-based fall detection systems have been actively studied over the past decade due to their low privacy concerns and convenient nonwearable nature. In particular, WiFi channel state information (CSI)-based solutions offer the additional benefit of requiring no specific hardware due to the widespread deployment of WiFi. However, WiFi CSI-based systems typically suffer from severe performance degradation in cross-domain scenarios, non-line-of-sight (NLOS) and/or through-the-wall (TTW) environments. For successful real-world deployment, a fall detection system must be able to provide reasonably good performance even in such poor conditions. In this article, we propose a new WiFi CSI amplitude-based fall detection system (called AmFall) that effectively addresses the aforementioned challenges. To cope with NLOS/TTW situations, AmFall selectively uses the subcarriers and principal component containing valuable information. Considering that the continuous wavelet transform (CWT) is suitable for time-frequency analysis of nonstationary signals, such as those generated during falls, AmFall generates a scalogram by applying CWT to the resulting CSI signal with a novel denoising algorithm and extracts the speed information for segmentation. The segmented scalograms are fed into a deep learning (DL)-based classifier. In this article, we also suggest a simple yet effective dataset augmentation method to generate multiple scalograms from a single CSI observation. The proposed system, implemented on commercial WiFi devices, achieves a high fall detection accuracy of 95.28% to 99.67% in TTW and cross-domain environments, outperforming state-of-the-art WiFi-based methods.**

***Index Terms*—Channel state information (CSI), convolutional neural network (CNN), fall detection, scalogram, WiFi sensing.**

I. Introduction

FALLS are the second leading cause of unintentional injury deaths worldwide [1]. A fall is defined as an event in which a person inadvertently falls to the ground, floor, or other low places. Older people and children are especially prone to falls, often with higher severity of injury.

Various systems have been proposed for fall detection, which can be broadly grouped into two categories: 1) wearable and 2) nonwearable. Wearable systems [2], [3], [4] rely on specialized devices with various sensors that are attached to the body or carried by the user. However, these systems can detect falls only when the devices are worn or held, limiting their applicability and practicality, especially for the elderly individuals who may not consistently carry such devices.

Nonwearable solutions include camera-, infrared-, acoustic-, vibration-, and radio frequency (RF)-based systems. Camera-based systems [5], [6] can be accurate in controlled environments but raise privacy concerns and detect falls only when captured on camera. Infrared-based systems [7], [8] detect falls using thermal data but face challenges, such as sensor range limitations, interference from ambient temperatures, and privacy concerns. Acoustic systems [9] using inaudible signals are based on the Doppler effect but may cause discomfort and have reduced accuracy in non-line-of-sight (NLOS) scenarios. Vibration-based systems [10] using geophone sensors are affected by floor types and often produce false alarms. RF-based systems detect falls by analyzing the reflected radio-signals from a moving or falling person, reducing privacy concerns.

To leverage the advantages of RF-based sensing (i.e., nonwearable, low privacy concern), many fall detection systems using various radars have been proposed: the pulse Doppler radar (5.8 GHz) [11], the frequency modulated continuous wave (FMCW) Doppler radar (25 GHz, 58 — 63 GHz, and 60 — 64 GHz) [12], [13], [14], the continuous wave (CW) Doppler radar (24 GHz) [15], and the ultra-wideband (UWB) radar [16], [17]. However, these radar-based systems require specialized devices, limiting their applications due to the increased costs.

WiFi channel state information (CSI) sensing also has gained great attention due to the widespread deployment of WiFi devices in addition to the merits of RF-based sensing. WiFi CSI represents how signals propagate from a transmitter (Tx) to a receiver (Rx) over multiple paths. By analyzing CSI, human activities or motions can be inferred. WiFi CSI-based solutions offer benefits of being nonwearable, requiring no specific hardware, being ubiquitously deployable, and adhering to standardization, but there are still some challenges in real-world deployment.

1) Most real-world falls likely occur under strong NLOS or through-the-wall (TTW) conditions.[1] The increased

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[1]In this article, *strong NLOS* refers to cases where there is no direct path between a person and any of the transceivers, *weak NLOS* means no direct path between a person and one of the transceivers, and LOS denotes direct paths between a person and all transceivers.

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noise under these conditions requires advanced preprocessing to filter out noise while retaining useful information.

2) The fall detection performance can be degraded in cross-domain scenarios, where the training and testing environments differ from each other.[2]
3) An ample training dataset is crucial for fall detection classifiers. However, collecting a genuine dataset is challenging due to safety risks. A generative machine learning model [18] can address this, but it requires sophisticated tuning and specific processing to be effective.
4) Effective segmentation of time-series CSI data is essential to isolate individual activities, not only to enhance classification performance but also to ensure success in real-world deployment.
5) Considering the processing capabilities and limitations of the actual deployment environment, the appropriate solutions need to be selectively deployed.

In this article, we propose a WiFi CSI *am*plitude-based *fall* detection system (AmFall, hereafter) that effectively addresses the challenges mentioned above. We first outline the overall operation of AmFall and then describe how it handles the challenges.

In AmFall, a training dataset is expanded using the following method. We consider that the physical channel in orthogonal frequency division multiplexing (OFDM)-based WiFi systems (e.g., IEEE 802.11a, IEEE 802.11n) is composed of many subcarriers. For a given time period, when the received CSI values of a subcarrier signal are called a *CSI stream* (i.e., time-series CSI samples) and the received CSI values of all subcarriers at a single WiFi receiver are called a *CSI observation*, a CSI observation consists of as many CSI streams as the number of subcarriers. When a CSI observation is split into multiple distinct subgroups and the CSI streams of each subgroup is used as one input data, multiple inputs are generated from a single CSI observation and as a result, the dataset can be expanded. For each input data, some CSI streams containing more useful information (not its all streams) are selectively used. The system applies the principal component analysis (PCA) technique to these selected CSI streams and chooses only effective principal components (PCs), which are ultimately combined into a single CSI stream. This process not only reduces the data size but also has a denoising effect to reduce the noise. Considering that a continuous wavelet transform (CWT) is suitable for the time-frequency analysis of nonstationary signals, the system creates a scalogram from the combined stream, removes residual noise from the scalogram, and extracts speed information from the denoised scalogram for segmentation. The proposed system finally checks whether a fall activity has occurred using the segmented scalogram.

Now, we describe how the five challenges are tackled in our system. For the first challenge (severe noisy condition), the system achieves a denoising effect while retaining useful information by selecting CSI streams and PCs based on their sensitivity and eliminating the residual noise from the generated scalogram. The second challenge (cross-domain robustness) is addressed using appropriately denoised scalograms, which exhibit similar patterns for the same activity across different environments (e.g., sharp horn patterns for falls). In addition, since the speed information extracted from the denoised scalograms is independent of environments, this cross-domain issue is further mitigated. The third challenge (creating an ample dataset) is handled by splitting a single CSI observation into multiple subgroups of CSI streams, each of which is used as an input. The fourth challenge (segmentation) is effectively addressed by finding the activity period from the scalogram based on the speed information. Our approach for handling the fifth issue (deployment condition) is to detect fall activities either using the extracted speed information alone or applying deep learning (DL) models of varying depth to the scalogram images, depending on deployment conditions.

By effectively handling the challenges, AmFall achieves significantly higher accuracy compared to existing state-of-the-art fall detection systems [19], [20], [21], [22], in cross domain scenarios with CSI data measured in strong NLOS and TTW environments.

The main contributions of this article are as follows.

1) We develop a fall detection scheme with high accuracy and short inference time under TTW conditions and cross-domain scenarios.
2) A simple yet effective dataset augmentation method for WiFi-based sensing data is suggested.
3) We design a CSI stream sensitivity metric to evaluate how much activity-related information a CSI stream contains and develop a novel preprocessing method that selects useful CSI streams and PCs based on their sensitivity.
4) An efficient denoising method, which removes fall-independent components and noise while retaining fall-related information in the scalogram, is proposed. In addition, a new segmentation method to effectively identify the actual activity period using environment-independent speed information extracted from the denoised scalogram is developed.
5) We suggest multiple options of AmFall depending on deployment conditions, such as processing capability or environmental limitations: speed information-based fall detection, lightweight DL model-based fall detection, or heavy DL model-based fall detection.

The remainder of this article is organized as follows. Section II reviews related works. Section III describes the details of the proposed method. In Section IV, we evaluate the fall detection performance of the proposed system. This article is concluded with Section V.

II. Related Works

Since CSI toolkits using commercial WiFi chips [23], [24] have been released, numerous studies have been conducted with these toolkits for various applications, such as fall detection, gesture recognition, gait analysis, and more. This

[2]For clarity, the term *domain* refers to the *environment* in this article.

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section reviews existing WiFi CSI-based works, primarily focusing on fall detection.

WiFall [25] used CSI amplitude variation patterns in the time domain to detect falls from other human activities and employed a one-class support vector machine (SVM) and random forests as fall classifiers. The authors applied the singular value decomposition (SVD), taking components with the three largest eigenvalues to reduce processing time. Their method targeted line-of-sight (LOS) scenarios from an intradomain perspective, where training and test data were collected in the same environment. RT-Fall [26] leveraged CSI phase information by using phase differences between two Rx antennas to remove random phase offsets and extracted features from CSI amplitude and phase difference. An SVM classifier was used to detect falls and its performance was evaluated under LOS and weak NLOS conditions. Both WiFall and RT-Fall are environment-dependent, meaning that the extracted features in one environment may not perform well in different environments.

To tackle environmental dependency, FallDeFi [27] selected PCs by applying PCA to denoised CSI amplitudes and then extracted time-frequency features from short-time Fourier transform (STFT) spectrograms of the selected PCs. The authors considered various fall types in both LOS and NLOS scenarios, but the fall detection rate was greatly decreased in cross-domain. TL-Fall [28] applied PCA to CSI amplitudes and then extracted features by applying discrete wavelet transform (DWT) to the second PC. To mitigate environmental dependency, TL-Fall proposed a SVM-based transfer learning method that requires a small set of labeled data for retraining in a new environment. However, to achieve a specific level of performance, a considerable amount of training data was still needed for the target environment. Moreover, the tests were conducted only in LOS scenarios.

WiSpeed [19] demonstrated a remarkable achievement in speed estimation using WiFi CSI. It extracted speed information from a moving object by treating each scatter as a virtual Tx antenna, applying the statistical theory of electromagnetic waves to the power of CSI, and differentiating falls from four daily activities by thresholding. WiSpeed achieved comparable performance in strong NLOS but its performance was evaluated with a small test dataset in a single environment for fall detection. DeFall [29] leveraged the concept of WiSpeed to extract speed information. After creating a typical human fall template based on dynamic time warping (DTW) using the extracted speed, it classified falls by comparing the patterns of real-time speed/acceleration estimates with the representative template. Based on WiSpeed, FallAware [30] computed the differential of the auto-correlation function of CSI power and employed a lightweight LSTM-based neural network for classification instead of directly using the extracted speed information. FallDar [18] adopted a similar approach to WiSpeed and DeFall in estimating the speed of a moving object. It proposed a generative deep neural network (DNN) to generate a large amount of synthesized data from a small set of real samples and utilized an adversarial learning-based DNN classifier to extract user-independent fall features. Tests were conducted primarily under LOS scenarios, along with weak NLOS conditions. WiSpeed, DeFall, FallAware, and FallDar all rely on speed information as a key feature, because the speed generated by the human body is independent of the environment. The underlying theory behind these systems assumes rich scattering environments, such as chambers, offices, or homes. Consequently, their performance may be degraded in poor scattering environments like large halls or in strong LOS conditions.

Nakamura et al. [20] applied PCA to CSI amplitudes, selected PCs until the cumulative ratio of their eigenvalues exceeded a threshold, and then combined the spectrograms generated by applying the STFT on the selected PCs. The combined spectrogram images were segmented and fed into convolutional neural network (CNN) models, primarily pretrained ResNet-34, for classification. Evaluation was performed under LOS scenarios with a limited range of fall types. Chu et al. [21] generated an image by mapping three Rx antenna inputs to RGB channels and classified it using a pretrained EfficientNet-B0 with minor customizations. Tests were primarily conducted in LOS environments, with one NLOS setting. This method performed well in intradomain scenarios but showed poor performance in cross-domain scenarios. Furthermore, three Rx antennas were required for image generation. DapFall [22] transformed CSI amplitude into a dynamic amplitude probability density for 3-D visualization and classified falls using the R2+1D CNN model with data augmentation to overcome the limited fall data.

In addition to the works mentioned above, several other studies have explored fall detection using different technologies. AFall [31] used the change of angle of arrival (AoA) over time exceeding a threshold as a fall detection criterion. It provided comparable performance in LOS scenarios but had significant performance degradation in NLOS. In Fall-Attention [32], the smoothed STFT spectrograms, derived from the corrected phase difference of CSI, were fed into a DL classifier that utilized a recurrent neural network (RNN)-based sequence encoder followed by attention-based sequence embedding. This approach was evaluated in LOS scenarios with a limited range of fall types.

While experimental scenarios of many existing works mainly consider LOS or weak NLOS environments, the actual deployment conditions are likely to be strong NLOS or TTW. Moreover, they still provide low performance in cross-domain scenarios. In this article, we propose a fall detection scheme with high accuracy in cross-domain and TTW conditions.

III. Methodology

The proposed fall detection system utilizes scalograms derived from WiFi CSI data collected by commodity WiFi devices. Fall detection is done using either DL models or speed information extracted from scalograms. The process follows four stages in Fig. 1: 1) dataset augmentation; 2) CSI preprocessing; 3) scalogram processing; and 4) classification.

A. Dataset Augmentation

We first explain a raw WiFi CSI observation, $\mathbf{H}_{\text{raw}}$, and describe how an input dataset is augmented using $\mathbf{H}_{\text{raw}}$.

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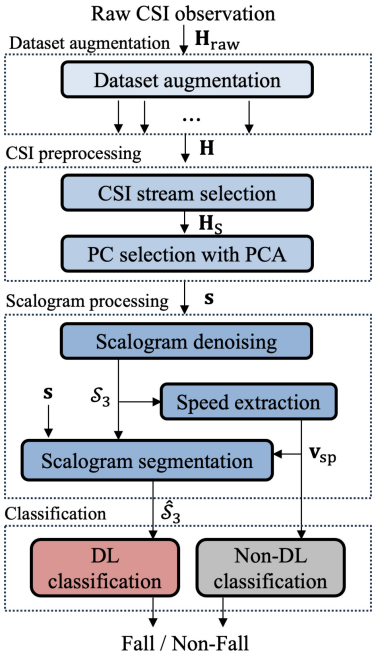


Fig. 1. Workflow of the proposed system.

1) CSI Data Representation: CSI describes how wireless signals propagate from a Tx to an Rx over multiple paths at a given carrier frequency. It encapsulates the amplitude attenuation and phase shift of each multipath signal, which is modeled as the channel frequency response (CFR), as described in [33]

$$H(f;t) = \sum_p \alpha_p(t) e^{-j2\pi f \tau_p(t)} \tag{1}$$

where $\alpha_p(t)$ represents the signal attenuation along the pth path, and $-2\pi f\tau_p(t)$ corresponds to the phase shift caused by the propagation delay $\tau_p(t)$ at frequency f.

Due to multiple-input multiple-output (MIMO) and OFDM technologies in WiFi access points, a raw CSI observation collected by an Rx is represented by a matrix with dimensions $N \times L \times M \times K$, where N represents the number of time indices, L is the number of Tx antennas, M is the number of Rx antennas, and K is the number of subcarriers.

By letting $i = (l, m, k)$, where l represents the Tx antenna index, m is the Rx antenna index, and k is the subcarrier index, a CSI stream of a subcarrier corresponding to the index i, denoted by $\mathbf{h}_i$, is defined as

$$\mathbf{h}_i = \left[H(f_i;1), H(f_i;2), \ldots, H(f_i;N)\right]^{\text{T}} \tag{2}$$

where f_i is the carrier frequency corresponding to subcarrier k for the index i, and T denotes the transpose operator. When $N_H = L \cdot M \cdot K$, a raw CSI observation $\mathbf{H}_{\text{raw}}$ is represented as a 2-D matrix of $N \times N_H$

$$\mathbf{H}_{\text{raw}} = \left[\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{N_H}\right]. \tag{3}$$

2) Dataset Augmentation: In general, DL networks need a significant amount of training data, which can be time-consuming and resource-intensive to collect. Additionally, for fall detection, data collection poses risks to volunteers during the experiment. To overcome this, we suggest a method to generate multiple scalogram images from a single raw CSI observation, thereby expanding the dataset without additional real-world experiments.

The basic idea for dataset augmentation is that a few of $\mathbf{h}$'s (not all $\mathbf{h}$'s in $\mathbf{H}_{\text{raw}}$) can be used for scalogram image generation, without performance degradation. Let $\mathbf{H}$ denote an input data matrix $\mathbf{H}$, of which the columns are the CSI streams used to generate a single scalogram image. We make $\mathbf{H}$ by choosing some columns of $\mathbf{H}_{\text{raw}}$. Note that the number of selected $\mathbf{h}$'s should be sufficient to generate a scalogram containing useful information. With the restriction that $\mathbf{H}$ contains at least J CSI streams, the number of $\mathbf{H}$ that can be made from a single $\mathbf{H}_{\text{raw}}$ is $\sum_{j=J}^{N_H} \binom{N_H}{j}$ at maximum. However, since the scalogram images from two $\mathbf{H}$'s containing multiple identical $\mathbf{h}$'s may be similar to each other, it is likely to be desirable that the $\mathbf{H}$'s derived from a single $\mathbf{H}_{\text{raw}}$ are configured to share as few CSI streams as possible. As an example, three $\mathbf{H}$'s can be obtained by splitting the CSI matrix $\mathbf{H}_{\text{raw}}$ into three parts with the identical size along a column-axis. That is, for $\mathbf{H}_{\text{raw}} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{180}]$ composed of the CSI streams of 180 subcarriers, $\mathbf{H}_1 = [\mathbf{h}_1, \ldots, \mathbf{h}_{60}]$, $\mathbf{H}_2 = [\mathbf{h}_{61}, \ldots, \mathbf{h}_{120}]$, $\mathbf{H}_3 = [\mathbf{h}_{121}, \ldots, \mathbf{h}_{180}]$. Since the proposed data augmentation can be simply implemented through this kind of matrix reshaping operation, its computational overhead is expected to be negligible. The effectiveness of this approach is demonstrated through the experiments in Section IV.

It is important that $\mathbf{H}$ consists of genuine CSI streams rather than synthesized one. Since our method is not a way to augment data itself but increases the number of training data within the dataset, we refer to it as dataset augmentation in this article.

B. CSI Preprocessing

1) CSI Stream Selection: Since the CSI streams ($\mathbf{h}$'s) typically contain significant noise, denoising is crucial for extracting activity-related features. While filtering can remove nonhuman activity components, denoising within human activity-related components is more challenging due to the risk of losing useful information or distorting the data. Our approach is to select high-quality CSI streams rather than extensive filtering, as frequency ranges beyond human activity can be effectively removed during scalogram processing. According to our experiments, only a small number of well-chosen CSI streams are sufficient for fall detection.

We utilize a moving variance of CSI amplitude as a metric for signal sensitivity. The moving variance for a CSI stream $\mathbf{h}$ at time index n is defined as

$$\upsilon(n;\mathbf{h}) = \frac{1}{2W} \sum_{k=n-W}^{n+W} \left(\left|\mathbf{h}[k]\right| - \mu_{\mathbf{h}}(n)\right)^2 \tag{4}$$

where $|\mathbf{h}[k]|$ is the amplitude of $\mathbf{h}[k]$ (in this article, $|x|$ represents the amplitude of x) and $\mu_{\mathbf{h}}(n)$ is the mean of $|\mathbf{h}|$ within the window $[n-W, n+W]$. The window size is $2W+1$. The corresponding time-series moving variance vector for $\mathbf{h}$ is defined as

$$\boldsymbol{\upsilon}(\mathbf{h}) = \left[\upsilon(1;\mathbf{h}), \upsilon(2;\mathbf{h}), \ldots, \upsilon(N;\mathbf{h})\right]^{\text{T}}. \tag{5}$$

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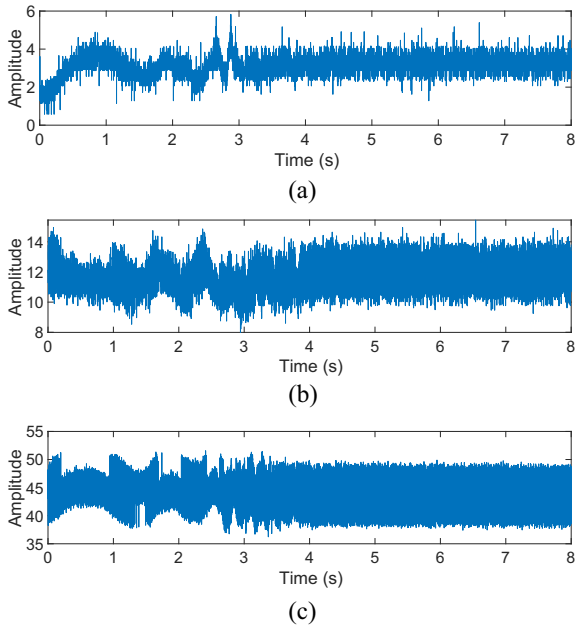


Fig. 2. Raw CSI streams with different q values. (a) Raw CSI stream with highest q. (b) Raw CSI stream with median q. (c) Raw CSI stream with lowest q.

Note that $\boldsymbol{\upsilon}(\mathbf{h})$ represents the strength of fluctuations caused by noise as well as moving objects in the sensing area. To mitigate the impact of noise, we normalize the maximum value with the mean, as follows:

$$q(\mathbf{h}) = \frac{\max(\boldsymbol{\upsilon}(\mathbf{h}))}{\text{mean}(\boldsymbol{\upsilon}(\mathbf{h}))}. \tag{6}$$

It is likely that a higher $q(\mathbf{h})$ indicates stronger fluctuation caused by moving objects, implying higher sensitivity to human activities. Fig. 2 illustrates that CSI streams with higher $q(\mathbf{h})$ contain more information for detecting activity, despite lower amplitude values. Thus, using $q(\mathbf{h})$, we prioritize CSI streams that can more accurately capture relevant activity while reducing the impact of noise.

Specifically, we first sort the CSI streams in $\mathbf{H}$ in descending order of their $q(\mathbf{h})$ values and consider only the top ℓ CSI streams, where ℓ is a hyperparameter. The ℓ CSI streams are reindexed from 1 to ℓ for convenience of description. The $q(\mathbf{h}_i)$ value of each $\mathbf{h}_i$ is normalized as follows:

$$Q(\mathbf{h}_i) = \frac{q(\mathbf{h}_i)}{\sum_{k=1}^{\ell} q(\mathbf{h}_k)}. \tag{7}$$

Second, among the top ℓ CSI streams, the CSI streams with Q value greater than or equal to $1/\ell$ are selected for further processing.

Let $\mathcal{N}_s$ be the index set of the selected CSI streams, and let n_s be the number of CSI streams in $\mathcal{N}_s$. The CSI data composed of n_s $\mathbf{h}$'s in $\mathcal{N}_s$, i.e., the CSI data after the CSI stream selection, is denoted by $\mathbf{H}_S$, which is a 2-D matrix of $N \times n_s$.

2) PCA and PC Selection: We use PCA for denoising and reducing the number of processed CSI streams, by selecting PCs that contain useful information with less noise. PCA is based on eigenvalue decomposition of covariance matrix $\mathbf{R}^{\mathbf{H}_S}$

$$\mathbf{R}^{\mathbf{H}_S}\mathbf{V} = \mathbf{V}\mathbf{D} \tag{8}$$

where $\mathbf{R}^{\mathbf{H}_S}$ is the $n_s \times n_s$ covariance matrix of $\mathbf{H}_S$, $\mathbf{D}$ is the diagonal matrix with eigenvalues, and $\mathbf{V}$ is the corresponding eigenvector matrix. When $\mathbf{v}_j$ denotes the jth eigenvector in $\mathbf{V}$, the jth PC is

$$\mathbf{p}_j = \mathbf{H}_S\mathbf{v}_j. \tag{9}$$

Selecting PCs is based on the eigenvalues in $\mathbf{D}$, since the eigenvalue represents the variance of each PC, indicating its significance. PCs with very small eigenvalues (referred to as minor PCs) can be discarded to reduce noise and data volume. However, even nonminor PCs may contain significant noise, especially when raw CSI data are collected in very noisy environments. Note that noisy nonminor PCs degrade performance. Therefore, it is important to evaluate the usefulness of each PC and select only those with valuable information, for further processing.

We use the Q metric in (7) to assess the usefulness of PC. First, the eigenvalue of each PC is normalized with the sum of all eigenvalues and then the candidate PCs with the normalized eigenvalue greater than a specified threshold are selected.[3] Next, we calculate the $Q(\mathbf{p})$ value of each candidate PC $\mathbf{p}$ by applying $\mathbf{p}$ instead of $\mathbf{h}$ to (4)–(7). When the number of candidate PCs is n_c, we finally select the PCs with the Q value greater than or equal to $1/(n_c - 1)$, among these n_c candidate PCs. If none of the PCs meet this criterion, the PC with the highest Q is selected.

C. Scalogram Processing

Human falls typically exhibit nonstationary characteristics, causing abrupt torso movements and sudden changes in WiFi multipath propagation lengths over short durations. The proposed fall detection scheme uses a CWT for time-frequency analysis, since it is suitable to nonstationary signal analysis [34]. According to [34], the CWT of a signal $s(t)$ using L1 normalization is

$$W(\nu, \tau; s) = \frac{1}{\sqrt{|\nu|}} \int s(t)\psi^*\left(\frac{t-\tau}{\nu}\right)dt \tag{10}$$

where ψ^* is the conjugate of the basic wavelet ψ, ν is the scaling factor and τ is the shifting factor.

The scalogram for the signal $s(t)$, CWT$(s(t))$, is represented as a 2-D matrix. Each row corresponds to its respective scaling factor, ν_i, which indicates a specific frequency band. Each column corresponds to its respective shifting factor, τ_j, representing a specific time index. Thus, each coefficient of the scalogram matrix is denoted as $W(\nu_i, \tau_j; s)$.

We regard the selected PCs as separate CSI signals, like in [20], [27], [35]. We can obtain a single scalogram by computing the scalogram of each selected PC and then summing the scalograms. However, when considering that $\sum_{j\in\mathcal{N}_{PC}} \text{CWT}(\mathbf{p}_j) = \text{CWT}(\sum_{j\in\mathcal{N}_{PC}} \mathbf{p}_j)$, where $\mathcal{N}_{\text{PC}}$ is the

[3]The threshold is adjusted adaptively so that the number of candidate PCs is not too small.

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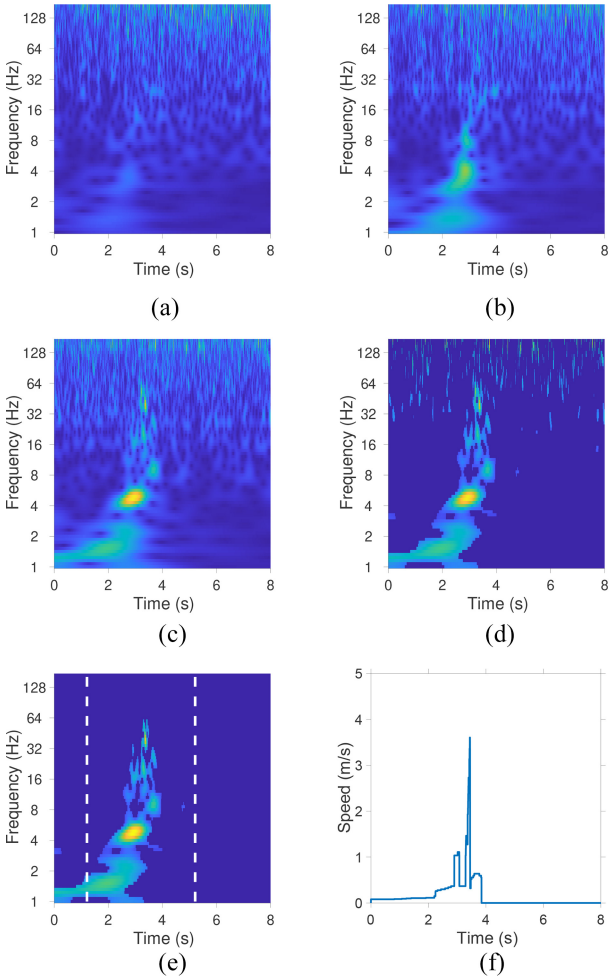


Fig. 3. Scalogram images in each preprocessing step. (a) $\mathcal{S}_0$ using $\mathbf{H}_{\text{raw}}$. (b) $\mathcal{S}_0$ using $\mathbf{H}_S$. (c) $\mathcal{S}_0(\mathbf{s})$ in (12). (d) $\mathcal{S}_1$ after general denoising. (e) $\mathcal{S}_3$. (f) Extracted speed information.

index set of the selected PCs, the computational overhead can be greatly reduced by first summing all selected PCs and then computing the scalogram for the sum.

The proposed method computes the normalized scalogram for a single CSI stream $\mathbf{s}$ that is the sum of the selected PCs

$$\mathbf{s} = \sum_{j \in \mathcal{N}_{\text{PC}}} \mathbf{p}_j. \tag{11}$$

$$\mathcal{S}_0(\mathbf{s}) = \left|\text{Normalize}\big(\text{CWT}(\mathbf{s})\big)\right| \tag{12}$$

where Normalize() means max normalization. In this article, the analytic Morse wavelet is applied to the frequency range from 1 to 170 Hz, which corresponds to the frequency range associated with human falls.

Fig. 3(a)–(c) illustrates $\mathcal{S}_0$ scalograms generated using a raw CSI matrix, $\mathbf{H}_S$ from the stream selection step, and $\mathbf{s}$ from the PC selection step. The effect of each preprocessing step can be visually observed by comparing the scalograms before and after the step.

Fig. 4 shows $\mathcal{S}_0(\mathbf{s})$ scalograms from two types of human activities done by the same volunteer in two different environments. It is observed from the figure that the same activities generate similar scalograms in different environments. In

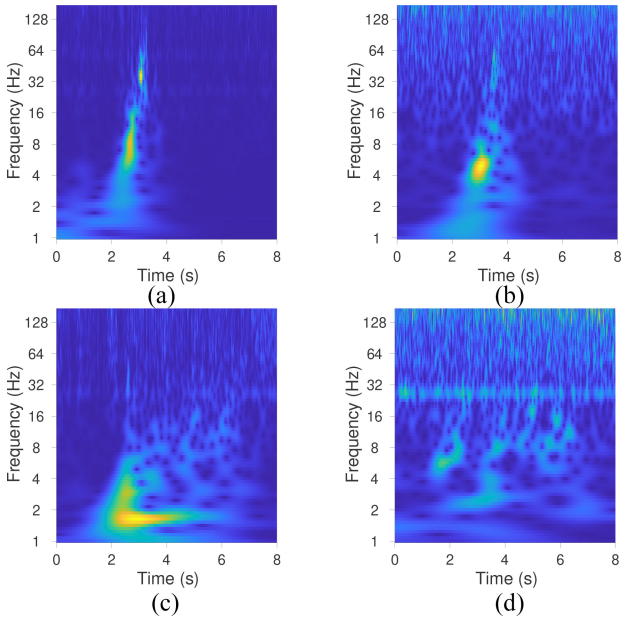


Fig. 4. $\mathcal{S}_0(\mathbf{s})$ scalograms of motions in two environments (A, B) by the same volunteer. (a) Falling in A. (b) Falling in B. (c) Lying down in A. (d) Lying down in B.

general, human movements within a sensing area induce propagation length variations in reflected signals from different body parts, which are captured by changes in $\alpha_p(t)$ and $e^{-j2\pi f \tau_p(t)}$ in (1), with the reflection strength proportional to $|\alpha_p(t)|$. These variations are mapped to frequency components through time-frequency analysis, whereby identical movements under consistent conditions yield similar frequency patterns. Although variations in subjects, locations, orientations, or measurement environments may introduce certain deviations in these patterns, fundamental similarities may inherently persist, thereby enabling classification through ML techniques. This perspective has been adopted in existing works [20], [27], [32]. Based on this insight, we leverage the scalograms to detect falls. Note that, even though the CSI stream and PC selection processes have the denoising effect, there may still be a significant amount of residual noise within $\mathbf{s}$. As a result, the scalogram $\mathcal{S}_0(\mathbf{s})$ also can contain considerable amount of noise, as shown in Figs. 3(c) and 4. Thus, scalogram denoising is needed.

Our scalogram denoising process consists of three steps: general denoising, which clears (i.e., sets to zero) small coefficient values, and vertical and horizontal denoising, which clear infeasible coefficient values with respect to frequency band indices and time indices, respectively.

1) General Denoising: General denoising is based on the intuition that the coefficients representing human activities typically have magnitudes greater than the average magnitude of the entire scalogram. Minor coefficients with relatively small magnitudes, which may result from noise or irrelevant activities, can be cleared through thresholding.

The challenge lies in determining a threshold adaptively to each scalogram. For noisy scalograms, the threshold needs to be high to effectively remove noise, whereas for cleaner scalograms, a lower threshold helps preserve useful information.

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Algorithm 1: General Denoising

Input: $q(\mathbf{s})$, $\mathcal{S}_0(\mathbf{s})$
Output: $\mathcal{S}_1$

$\mathcal{S}_1 \leftarrow \mathcal{S}_0(\mathbf{s})$;
$th_sc \leftarrow \min\Big(\big(\log_{10} q(\mathbf{s})\big)^{-1}, th_sc_{\max}\Big)$;
for *f in frequency-band range of* $\mathcal{S}_1$ **do**
  $th_f \leftarrow \text{mean}(\mathcal{S}_1(f,:)) \times th_sc$;
  $\mathcal{S}_1(f,n) \leftarrow 0, \ \forall n$ such that $\mathcal{S}_1(f,n) < th_f$;
return $\mathcal{S}_1$

Thus, we need to consider the quality and mean magnitude of scalogram in determining thresholds.

The thresholds are determined by the following two steps: the first step determines a threshold for the entire scalogram, *th_sc*, on a basis of $q(\mathbf{s})$, and the second step for fine tuning calculates a frequency threshold, *th_f*, for each frequency band by multiplying *th_sc* by the mean magnitude of the coefficients in that frequency band. In each frequency band, the scalogram coefficients with magnitude below the frequency threshold of the band are then cleared. We denote the resulting scalogram after general denoising by $\mathcal{S}_1$, exemplified in Fig. 3(d). Algorithm 1 summarizes our general denoising, where $th_sc_{\max}$ represents the upper bound of the first step threshold.

2) Vertical Denoising: Vertical denoising operates on the premise that the maximum Doppler frequency shift (DFS) induced by human torso movement is limited. The maximum DFS per time index, D, is approximately 0.17, given a sampling frequency of 1000 Hz in our experiments. This value is derived from the estimated maximum DFS caused by the free fall of a human torso, which is approximately 170 Hz. When the DFS change exceeds D, it can be inferred that this change does not originate from the human torso.

At a given time index, vertical denoising clears coefficients of frequency bands being unreachable from the previous highest frequency band. For instance, if the highest frequency band at time step n is f, applying the threshold D, the feasible frequency bands at time $(n+1)$ should be less than $(f + D)$. Then, any nonzero coefficients in the frequency bands greater than $(f + D)$ at time $(n + 1)$ can be regarded as noise. However, since CSI measurements may be inaccurate, we need to relax the decision criteria for infeasible coefficients rather than simply applying D, to avoid discarding potentially valuable data.

The proposed vertical denoising process is outlined in Algorithm 2. Specifically, at each time index, the frequency bands with a nonzero coefficient are compared with the highest frequency band at the previous time index (denoted by $f_{\text{p_max}}$), in descending order of frequency. The nonzero coefficient feasibility of each frequency band f_i is determined according to the following criteria.

1) A nonzero coefficient at f_i is feasible if $f_i \le f_{\text{p_max}}$.
2) Even if $f_i > f_{\text{p_max}}$, a nonzero coefficient at f_i is feasible if the following two conditions are satisfied. One checks whether $(f_i - f_{\text{p_max}}) \le \Delta_D$, where Δ_D is a predefined parameter representing an allowable maximum frequency change per time index. The other is to calculate the number of time indices required to reach f_i from $f_{\text{p_max}}$ and check whether the average of the highest frequency bands for this duration is close to $f_{\text{p_max}}$. That is, letting $d = (f_i - f_{\text{p_max}})/D$, this condition checks if $(\sum_{j=n-d}^{n-1} \max \mathcal{F}(:,j))/d \approx f_{\text{p_max}}$, where $\mathcal{F}$ is the matrix that contains the corresponding f_i values for the nonzero coefficients of $\mathcal{S}_1$.

Algorithm 2: Vertical Denoising

Input: $\mathcal{S}_1$: Scalogram from Algorithm 1
Output: $\mathcal{S}_2$

$$\mathcal{F}(i,n) \leftarrow \begin{cases} 0, & \text{if } \mathcal{S}_1(i,n) = 0 \\ f_i & \text{otherwise} \end{cases}, \ \forall i,n;$$
$f_{\text{p_max}} \leftarrow 0$;
for *n in time range of* $\mathcal{S}_1$ **do**
  while $f_i = \max \mathcal{F}(:,n)$ **do**
    if $f_i \le f_{\text{p_max}}$ **then**
      break;
    $d \leftarrow \frac{f_i - f_{\text{p_max}}}{D}$;
    if $(f_i - f_{\text{p_max}} \le \Delta_D)$ & $\Big(\big(\sum_{j=n-d}^{n-1} \max \mathcal{F}(:,j)\big)/d \approx f_{\text{p_max}}\Big)$ **then**
      break;
    $\mathcal{F}(i,n) \leftarrow 0$;
  $f_{\text{p_max}} \leftarrow f_i$;
$$\mathcal{S}_2(i,n) \leftarrow \begin{cases} 0, & \text{if } \mathcal{F}(i,n) = 0 \\ \mathcal{S}_1(i,n), & \text{otherwise} \end{cases}, \ \forall i,n;$$
return $\mathcal{S}_2$

Note that if a coefficient of f_i is feasible at a time index, all nonzero coefficients in lower frequency bands than f_i at that time index are feasible.

All components deemed infeasible are cleared because they are considered noise or are caused by nonhuman movement.

3) Horizontal Denoising: Horizontal denoising is based on the idea that nonzero coefficients within a frequency band appear consecutively for a certain period if they are caused by human activities. This implies that signal components reflecting activities such as human motion persist over time, distinguishing them from noise which is typically more sporadic. Let us infer this persisting time.

Fall activity in the sensing area is reflected in DFS, $f^{\mathbb{D}} = \text{v}/\lambda$ (ignoring the sign), where λ is the wavelength of the carrier frequency, and v is the velocity corresponding to the change in propagation length. Then, the DFS difference over time interval Δt is $(\Delta f^{\mathbb{D}}/\Delta t) = (1/\lambda)(\Delta \text{v}/\Delta t) = a/\lambda$, where a is the velocity change during Δt, i.e., acceleration. Since a is expected to be proportional to the gravitational acceleration g under the assumption of free fall, we set $a = \kappa \cdot g$, where a proportional constant κ is our hyperparameter. Then, Δt is calculated by $(\Delta f^{\mathbb{D}} \lambda/\kappa g)$ and is proportional to $\Delta f^{\mathbb{D}}$. By considering $\Delta f^{\mathbb{D}}$ as the bandwidth of each frequency band, Δt for each frequency band can be calculated. If nonzero

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Algorithm 3: Horizontal Denoising

Input: $\mathcal{S}_2$: Scalogram from Algorithm 2
Output: $\mathcal{S}_3$

$\mathcal{S}_3 \leftarrow \mathcal{S}_2$;
for f *in frequency band range of* $\mathcal{S}_3$ **do**
  $n \leftarrow 1$;
  while $n < N$ **do**
    $m \leftarrow$ Number of consecutive nonzero coefficients of $\mathcal{S}_3$ from an index n at f;
    if $0 < m < \Delta n(f)$ **then**
      $\mathcal{S}_3(f, n : n + m - 1) \leftarrow 0$;
    $n \leftarrow n + m + 1$;
return $\mathcal{S}_3$

coefficients at a frequency band do not persist during Δt, these coefficients can be regarded as unrelated to a human fall.

We summarize this horizontal denoising process in Algorithm 3. In the algorithm, Δt is converted to the corresponding number of time indices by dividing it by the interval between two consecutive time indices, T_s. Note that the higher (lower) frequency band in scalogram has the wider (narrower) bandwidth. Let $BW(f_i)$ denote the bandwidth of a frequency band f_i, and let λ_c represent the wavelength of the center frequency in the entire set of OFDM subcarriers. When $\Delta n(f_i)$ denotes the number of time indices representing the persisting period of nonzero coefficients at f_i, $\Delta n(f_i) = (BW(f_i)\ \lambda_c)/(\kappa g T_s)$. N is the total number of time indices in scalogram.

Fig. 3(e) illustrates the resulting scalogram, $\mathcal{S}_3$, after vertical and horizontal denoising. It is noted that a fall involves complex movements such as not only the torso but also the limbs and head, and nontorso motions also may provide valuable information for classification. Taking this into account, we design $\mathcal{S}_3$ to include nontorso movements by appropriately setting denoising parameters.

D. Segmentation and Speed Extraction

Segmentation is the process of trimming the scalogram to retain more useful information by discarding unnecessary parts. We leverage the following moving sum of the moving variances for the combined stream $\mathbf{s}$ of (11) to specify the period of human activity

$$\Lambda(n; \mathbf{s}) = \sum_{k=n-W_m}^{n+W_m} \upsilon(k; \mathbf{s}), \quad \text{for} \quad n = 1, \ldots, N \tag{13}$$

where the window size is $2W_m + 1$ and k is restricted from 1 to N. Then, a prominent peak of the moving sum is regarded as the center of an activity period. Since falls typically occur within 2 to 3 s, we take the segmentation window of 4 s. When the peak of moving sum is at time index n_{fall}, $\mathcal{S}_3$ is segmented at $[n_{\text{fall}} - 2/T_s,\ n_{\text{fall}} + 2/T_s]$, where $2/T_s$ is the number of time indices during 2 s.

On the other hand, since the prominent peak can result from not only falls but also unwanted activities or impulse noise, multiple peaks may appear, which occasionally occur in our datasets. It is important to find the peak most associated with fall activity for correct segmentation. Our approach for identifying the most relevant peak, i.e., segmentation window center, is as follows. We first clear the nonzero coefficients smaller than the average magnitude of all nonzero coefficients in $\mathcal{S}_3$. Next, we feed the resulting scalogram into the vertical denoising algorithm and extract the highest nonzero frequency band at each time index, denoted as a time-series vector $\mathbf{f}_{\text{high}}$. Finally, we get a time index with the highest frequency band by examining $\mathbf{f}_{\text{high}}$ only around the peaks and determine it as the center of the fall activity period. We denote the segmented scalogram as $\hat{\mathcal{S}}_3$.

In addition, we estimate the speed vector $\mathbf{v}_{\text{sp}}$ over time using $\mathbf{f}_{\text{high}}$

$$\mathbf{v}_{\text{sp}} = \mathbf{f}_{\text{high}} \lambda_c. \tag{14}$$

Note that the role of our segmentation process is twofold: fulfilling accurate segmentation aligned with the actual period of fall activity and providing valuable speed information for further analysis or classification task. The segmentation result is illustrated in Fig. 3(e) with white dotted lines, and Fig. 3(f) depicts the estimated speed over time.

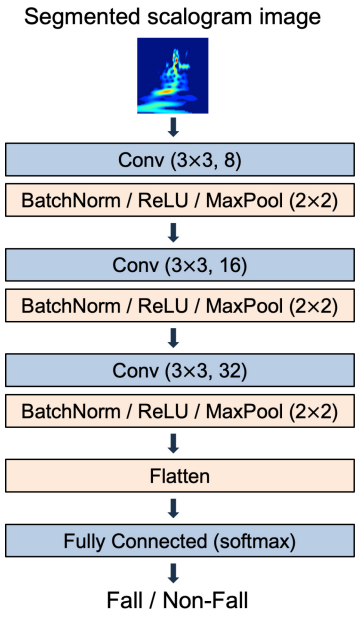


Fig. 5. Lightweight CNN model.

E. Classification

Our method offers two ways to detect a fall. One is by using the estimated speed $\mathbf{v}_{\text{sp}}$, and the other is through DL classification using an image from $\hat{\mathcal{S}}_3$, the segmented $\mathcal{S}_3$.

For DL classification, CNN is chosen due to its widespread use in image recognition, object detection, and time-series classification. We design a lightweight CNN model with three layers, referred to as LW-CNN, as shown in Fig. 5. The model consists of three 2-D convolution layers with 8, 16, and 32 filters, each of size 3×3. Each convolution layer is followed by a batch normalization layer, a ReLU activation layer, and

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a 2-D max-pooling layer. Finally, the model includes a fully connected layer with two outputs, followed by a softmax layer.

Besides LW-CNN, we evaluate the residual networks with 18 layers (ResNet-18) and 34 layers (ResNet-34) [36], which are chosen because they have similar architectures to each other but different depths. This is expected to allow us to assess the impact of our preprocessing method on model complexity. These two CNN models for comparison are used without customization, except for the final layer, which is modified for binary classification, and with pretrained parameters.

Our approach offers advantages in real-world deployment scenarios, by providing three options as follows.

1) If deployment environments lack DL computing capabilities, the extracted speed information can be used for fall detection without requiring DL processing.
2) When DL computing capabilities are limited, the lightweight LW-CNN model can be adopted.
3) When sufficient computing capabilities are available, deeper CNN models can be employed as needed.

IV. Evaluation

This section presents the experimental evaluations. Section IV-B covers the performance of the proposed method, Section IV-C describes the performance comparison results with other schemes, and Section IV-E presents the classification results based on the estimated speed without using DL technology.

A. Experimental Setup

1) Implementation: For the experiments, two desktops were used: one for Tx with two antennas and another for Rx with three antennas. Both desktops were equipped with an Intel WiFi Link 5300 card and the Linux 802.11n CSI tool [23]. CSI data was collected in monitor mode using the tool with the 5 GHz band, a bandwidth of 20 MHz, and a sampling rate of 1000 Hz for 8 s. The tool provides 30 subcarriers for each Tx-Rx antenna link pair, resulting in a total of 180 CSI data streams ($2 \times 3 \times 30 = 180$).

For CSI data analysis and preprocessing, a desktop with an Intel i5-10400 CPU and 64 GB of RAM was used, with the software implemented in MATLAB R2024a.

DL training and testing were performed using PyTorch [37] on Google Colab [38]. The default DL settings include the Adam optimizer with a weight decay of 0.0001, a cross-entropy loss function, a multistep learning rate scheduler (step size of 10 and gamma of 0.1), an initial learning rate of 0.0001, a batch size of 50, and 40 epochs. For DL-based classification, training and testing were conducted 15 times for each test case. In each session, the dataset was randomly shuffled, with 20% reserved for testing. When using the augmented dataset for training, the nonaugmented dataset was first split into training and testing sets, and the augmented training data were derived from the training set. The scalogram image for DL classification was represented in RGB format, rescaled to the range $[0, 255]$ with a 256-color map, and then resized to 224×224. These settings were applied to all DL evaluations.

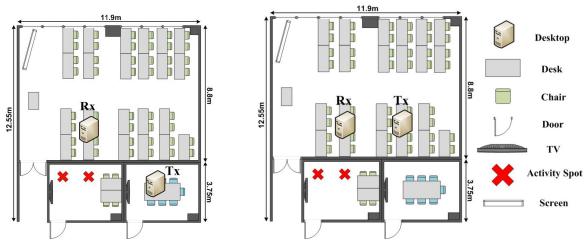


Fig. 6. Data collection environments. (a) Domain A. (b) Domain B.

TABLE I
Dataset Description

Dataset	Description
A	Single environment (Domain A), 5 volunteers, 300 falls (forward, backward, lateral), 300 nonfalls (sitting down, standing up, sitting on the floor, standing up from the floor, lying down, walking)
B	Single environment (Domain B), 3 volunteers, 180 falls (same as dataset A), 180 nonfalls (same as dataset A)
C [27]	Five environments, 3 volunteers, 304 falls (forward, backward, lateral, on-position), 347 nonfalls (walking, jumping, no event, random events, bending and picking up, sitting down, standing up)
D [39]	Three environments, 10 volunteers per environment, 400 falls (falling from a sitting position, falling from a standing position) per environment, 600 nonfalls (walking and turning, picking up a pen from the ground, sitting down and standing up) per environment

2) Dataset: We collected CSI data in two different environments (domain A and domain B), as shown in Fig. 6, both of which are TTW conditions. Five male volunteers (ages 20–30, heights 1.65–1.76 m) performed the activities in domain A and three of them repeated the activities in domain B. Each volunteer performed ten repetitions per spot for each fall type and five for each nonfall type. This resulted in 300 falls and 300 nonfalls in domain A, referred to as dataset A, and 180 falls and 180 nonfalls in domain B, referred to as dataset B. Each activity was recorded for 8 s, with a mattress for safety during falls.

In addition to the datasets (A, B) that we collected directly, two third-party datasets (C, D) are used for performance evaluation. Table I describes the details of these four datasets.

3) Performance Metrics and Data Representations: Four performance evaluation metrics are considered: 1) accuracy; 2) precision; 3) recall; and 4) F1 score. Accuracy represents the ratio of correctly classified falls and nonfalls to the total dataset. Precision indicates the ratio of correctly classified falls to all instances classified as falls. Recall measures the ratio of correctly classified falls to all actual falls. The F1 score is the harmonic mean of precision and recall.

Based on the dataset augmentation method in Section III-A, we summarize the augmentation types used in the experiments (1x means no augmentation), in Table II. It is noted that the augmented dataset is used for training only. For simplicity, A/B refers to training with the 1x augmented dataset from the dataset *A* and testing with the dataset B. B(12)/A indicates training with the 12x augmented dataset from the dataset *B*

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TABLE II
DATASET AUGMENTATION SETTINGS USED IN EXPERIMENT

Dataset	Augmentation Type	Descriptions (Subcarriers of Iput CSI Matrix)
A,B	1x	{1, 2, 3, ..., 180}
	6x	{1, ..., 30}, {31, ..., 60}, {61, ..., 90}, {91, ..., 120}, {121, ..., 150}, {151, ..., 180}
	12x	6x, {1, 7, 13, ..., 175}, {2, 8, 14, ..., 176}, {3, 9, 15, ..., 177}, {4, 10, 16, ..., 178}, {5, 11, 17, ..., 179}, {6, 12, 18, ..., 180}
C	1x	{1, 2, 3, ..., 60}
	6x†	{1, ..., 30}, {31, ..., 60}, {1, ..., 15}, {16, ..., 30}, {31, ..., 45}, {46, ..., 60}
D	1x	{1, 2, 3, ..., 90}
	3x	{1, ..., 30}, {31, ..., 60}, {61, ..., 90}

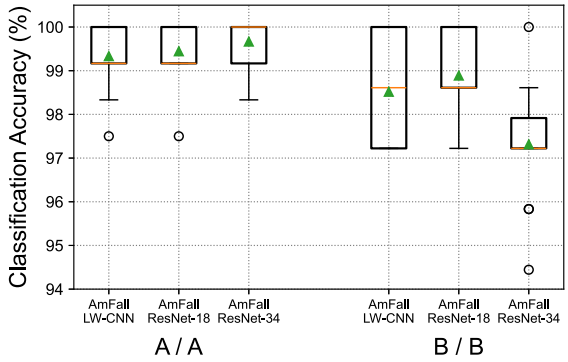


Fig. 7. Classification accuracy for the intradomains (A/A, B/B) with three AmFall CNN models.[4]

and testing with the dataset *A*. Similarly, AB(6)/C refers to training with the 6x augmented datasets from both datasets *A* and *B*, and testing with the dataset *C*.

B. Classification Results With CNNs

Three CNN models (i.e., LW-CNN, pretrained ResNet-18, pretrained ResNet-34) are applied to the same scalogram images generated from the preprocessing of AmFall. They are denoted by AmFall/LW-CNN, AmFall/ResNet-18, and AmFall/ResNet-34, respectively.

1) Intradomain: The intradomain evaluation means to training and testing on datasets collected in the same environment. Since our datasets *A* and *B* are from different domains, the intradomain evaluations are represented as A/A and B/B.

Fig. 7 shows the intradomain performance. Regardless of the CNN models or datasets used, all models achieve peak accuracies of 100% for both A/A and B/B, with average accuracies of over 99.3% for A/A and 97.3% for B/B. Actually, as can be expected from Fig. 4, the signal propagation environment of the dataset *B* is much noisier than that of the dataset *A*. However, the proposed system classifies falls with high accuracy even in the significantly noisy environment.

2) Cross-Domain: The cross-domain classification performance is denoted as A/B and B/A, where the training and testing datasets come from different domains. The 1x, 6x, and 12x augmentations are used for training, while only the 1x dataset is used for testing because augmentation is not necessary for testing. The classification performances are shown in Table III for the 1x training, Table IV for the 6x training, and Table V for the 12x training.

All AmFall CNN models achieve performance improvements through the dataset augmentation. While the performance of AmFall/LW-CNN is slightly improved, AmFall/ResNet-18 and AmFall/ResNet-34 exhibit much greater enhancements with the 6x and 12x augmentations, particularly in the A/B scenario. This is because ResNet-18 and ResNet-34 have significantly more parameters and therefore need more diverse training data to achieve better performance.

Our augmentation method effectively meets the data requirements for these complex CNN models. Additionally, even with the lightweight CNN model, the classification performance is comparable to those of much deeper models. This demonstrates that a lightweight CNN can be highly effective when the data are properly preprocessed.

3) Effect of CSI Processing and Scalogram Denoising: The preprocessing of AmFall consists of CSI preprocessing and scalogram denoising (see Fig. 1). To concentrate on preprocessing, dataset augmentation, and segmentation are not applied in this evaluation.

It is noted that the CWT is an essential operation in AmFall, because the DL-based classifiers use a scalogram image as input and the non-DL classifier exploits the speed information extracted from a scalogram. Thus, we assess the performance using the resulting scalogram image while progressively incorporating each processing step. Specifically, the scalograms are $\mathcal{S}_0(\sum_{i=1}^{N_H} |\mathbf{h}_i|)$ for no preprocessing (i.e., raw CSI), $\mathcal{S}_0(\sum_{i\in\mathcal{N}_S} |\mathbf{h}_i|)$ after the stream selection step, and $\mathcal{S}_0(\mathbf{s})$ after the PC selection step. In addition, $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$ are the scalogram images after general denoising, after vertical denoising, and after horizontal denoising, respectively. These images are fed into the LW-CNN model, because ResNet-18 and ResNet-34 may suffer from performance degradation due to insufficient training data in scenarios, such as A/B without the dataset augmentation (refer to Table III).

In Fig. 8, we can investigate the effectiveness of a specific preprocessing step by comparing the performances before and after the step. For example, the performance with the label "Raw CSI" demonstrates the effect of only the CWT, without preprocessing. The effect of the stream selection step can be evaluated by examining the performance difference between "Raw CSI" and "Stream Selection." We can observe in Fig. 8 that the performance is improved most significantly by selecting high-quality CSI streams and the performance gain from the PCA operation is also quite noticeable. The scalogram denoising has the effect of decreasing the deviation of the fall detection accuracy.

On the other hand, in Fig. 8, we can see a large performance difference between the A/B and B/A scenarios under the "Raw CSI" condition. This is because models trained in a better environment may not work well in worse environments. When the raw CSI data is used as input without any denoising, the model trained on dataset *A* containing less noise can perform

[4]Each quartile boxplot displays the median (Q2) as a line within the box spanning Q1 to Q3 (interquartile range, IQR). Whiskers extend up to 1.5×IQR from the quartiles, and outliers beyond this range are marked with circular symbols. A green triangle indicates the mean, if shown.

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TABLE III
CROSS-DOMAIN PERFORMANCE (%) WITH 1X (I.E., NONAUGMENTED) TRAINING DATASET

Model	B / A				A / B			
	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score
AmFall/LW-CNN	**97.33** (96.61)	**96.71** (95.52)	**98.00** (97.82)	**97.35** (96.65)	**93.61** (92.39)	**98.16** (98.08)	**88.89** (86.48)	**93.29** (91.89)
AmFall/ResNet-18	**96.83** (91.82)	**96.07** (86.62)	**97.67** (99.36)	**96.86** (92.47)	**81.94** (75.67)	**100.0** (100.0)	**63.89** (51.33)	**77.97** (67.71)
AmFall/ResNet-34	**96.83** (91.40)	**95.18** (85.92)	**98.67** (99.60)	**96.89** (92.16)	**85.83** (77.65)	**100.0** (99.88)	**71.67** (55.37)	**83.50** (70.85)

The bold numbers represent the highest accuracy achieved across 15 executions, while the numbers in parentheses indicate the corresponding mean values.

TABLE IV
CROSS-DOMAIN PERFORMANCE (%) WITH 6X TRAINING DATASET

Model	B(6) / A				A(6) / B			
	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score
AmFall/LW-CNN	**98.17** (97.10)	**97.69** (95.78)	**98.67** (98.56)	**98.18** (97.14)	**95.56** (94.11)	**98.81** (98.78)	**92.22** (89.33)	**95.40** (93.81)
AmFall/ResNet-18	**97.17** (95.67)	**94.92** (92.24)	**99.67** (99.78)	**97.24** (95.85)	**96.11** (94.87)	**100.0** (99.12)	**92.22** (90.56)	**95.95** (94.63)
AmFall/ResNet-34	**97.33** (94.98)	**95.22** (91.13)	**99.67** (99.78)	**97.39** (95.23)	**96.11** (94.87)	**98.26** (98.42)	**93.89** (91.22)	**96.02** (94.67)

TABLE V
CROSS-DOMAIN PERFORMANCE (%) WITH 12X TRAINING DATASET

Model	B(12) / A				A(12) / B			
	Accuracy	Precision	Recall	F1 score	Accuracy	Precision	Recall	F1 score
AmFall/LW-CNN	**98.17** (97.67)	**96.46** (96.95)	**100.0** (98.44)	**98.20** (97.69)	**95.28** (94.00)	**98.80** (99.59)	**91.67** (88.37)	**95.10** (93.64)
AmFall/ResNet-18	**99.00** (97.20)	**98.36** (94.85)	**99.67** (99.87)	**99.01** (97.28)	**96.11** (95.11)	**98.26** (99.12)	**93.89** (91.04)	**96.02** (94.90)
AmFall/ResNet-34	**99.67** (97.77)	**99.39** (95.88)	**100.0** (99.87)	**99.67** (97.82)	**95.28** (94.61)	**98.80** (99.15)	**91.67** (90.00)	**95.10** (94.35)

poorly when testing on dataset *B* with more noise, compared to the scenario B/A, i.e., training on dataset with more noise and testing on dataset with less noise. In the A/B scenario, the noise level of the raw test data is greatly reduced by the stream selection process which selects only the high-quality CSI streams with low noise level. As a result, the classification performance on A/B is significantly improved by the stream selection, and subsequent preprocessing steps also gradually remove residual noise within test data, leading to a noticeable improvement. Meanwhile, in the B/A scenario where the noise level of the raw test data is not high, the denoising effect of each preprocessing step is not dramatic, having minimal impact on performance improvement.

4) Evaluation Using Third Party Dataset: We evaluate the proposed method using two public datasets from [27] and [39], referred to as dataset *C* and dataset *D*, respectively.

The dataset *C* includes four types of falls (forward, backward, lateral, and on-position) and various nonfall activities (walking, jumping, no event, bending and picking up, sitting down, standing up, and random events). These activities were measured in five environments using an Intel WiFi Link 5300 card (one Tx with one antenna and one Rx with two antennas) and the Linux 802.11n CSI tool [23], for a duration of 10 s at a 1000 Hz sampling frequency in the 5.2 GHz band. Since the dataset *C* is relatively small and includes fewer antenna pairs compared to our datasets, we treat it as a single environment and expand it according to the 6x† augmentation in Table II. The dataset C and its augmented version are denoted as C and C(6†), respectively.

The dataset D consists of three environments. It includes two types of falls (falling from a sitting position and falling from a standing position) and three types of nonfall activities (walking and turning, sitting down and standing up, and picking up a pen from the ground). The duration of an

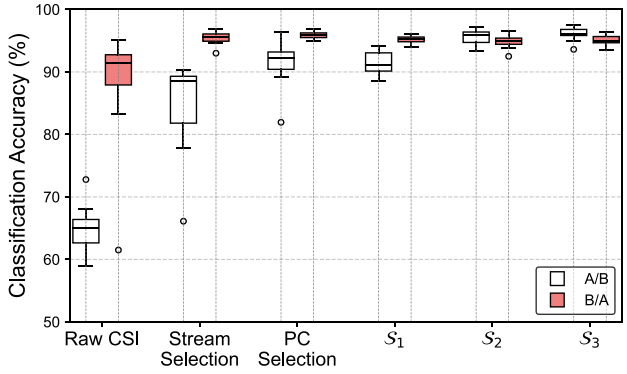


Fig. 8. Effect of each AmFall preprocessing on performance in A/B and B/A, evaluated using LW-CNN.

experiment varies from 4.5 to 16.4 s and can be subdivided into multiple activity segments. The dataset was collected using an Intel WiFi Link 5300 card (one transmitter with one antenna and one receiver with three antennas) and the Linux 802.11n CSI Tool [23], operating at a sampling rate of 320 Hz on the 2.4 GHz band. Each of ten volunteers performed the five experiments (two falls and three nonfalls) twenty times in each environment, resulting in 400 fall and 600 nonfall experiments per environment. We apply 3x augmentation (Table II) and refer to it as D(3).

Fig. 9 depicts the intradomain and cross-domain performances of AmFall using the third-party datasets. For cross-domain evaluation between our datasets and third-party datasets, we present only the results of C(6†)/AB and D(3)/AB because the ratio of unseen activities in AB/C and AB/D is high. Note that the unseen activity means an activity type seen only in testing but not present in training. When considering similar activities (e.g., *sitting down* and *sitting on the floor*) as the same activity, the ratio of unseen activity is approximately 8.3% for C(6†)/AB and D(3)/AB, whereas it is about 24.5%

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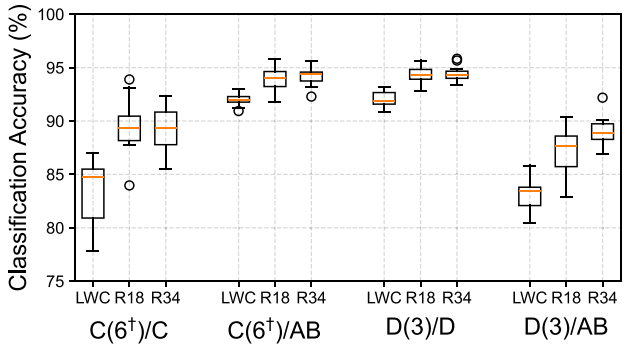


Fig. 9. Classification performance using third-party datasets *C* and *D*. LWC, R18, and R34 refer to AmFall/LW-CNN, AmFall/ResNet-18, and AmFall/ResNet-34, respectively.

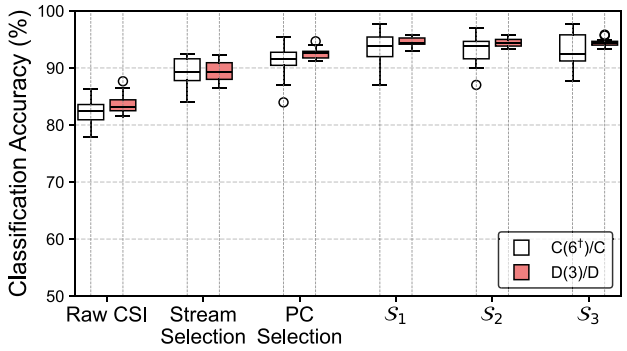


Fig. 10. Effect of each AmFall preprocessing on performance in C(6†)/C and D(3)/D, evaluated using ResNet-34.

for AB/C and 40% for D/AB.[5] Considering that the scenario with a high ratio of unseen activities may not be suitable for evaluating robustness to environmental changes, we conduct cross-domain experiments under C(6†)/AB and D(3)/AB.

In Fig. 9, for C(6†)/C, AmFall/ResNet-18 and AmFall/ResNet-34 achieve accuracies of up to 96.95%, and 94.66%, respectively, while AmFall/LW-CNN achieves an accuracy of only 87.02%. Since the FallDeFi dataset includes more diverse nonfall activities collected in a wider range of environments, the deeper CNN models may have an advantage in classification compared to the lightweight LW-CNN. The performance for C(6†)/AB is higher and more stable than that for C(6†)/C. This difference may be due to the number of CSI streams (i.e., 60 in the dataset *C* versus 180 in the datasets A and B). Consequently, the *A* and *B* datasets provide a greater chance of obtaining cleaner testing scalogram images from the larger CSI stream pool.

The dataset *D* exhibits typical performance characteristics. It demonstrates higher and more stable performance on D(3)/D than on D(3)/AB. Furthermore, the varying durations in the dataset D make segmentation essential, and the effectiveness of segmentation is more obvious under D(3)/D.

Next, in Fig. 10, we examine the effect of each preprocessing step on the performance of AmFall/ResNet-34, evaluated using the third-party datasets. Since the CSI observations in the dataset C have a fixed duration of 10 s, segmentation is not applied in the C(6†)/C setting in order to avoid potential segmentation errors. Meanwhile, since the CSI observations in the dataset D have different durations depending on activities, to generate fixed-size CNN inputs, segmentation with duration of 4 s is applied to the scalogram outputs of each processing step. We can see in Fig. 10 that the performance is progressively improved with each successive preprocessing step in both of C(6†)/C and D(3)/D.

By examining Figs. 8 and 10 together, we can see that the performance contribution of each step in the preprocessing pipeline may vary depending on noise conditions, cross-domain scenarios, and datasets. This implies the need of the entire preprocessing pipeline for coping with diverse situations.

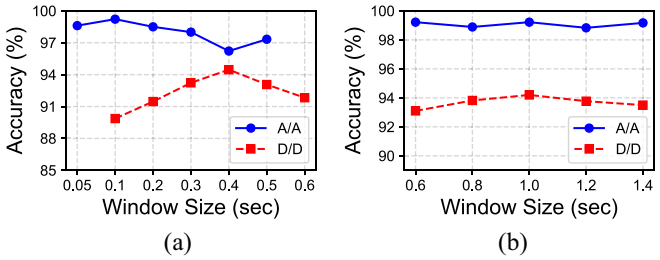


Fig. 11. Effect of window sizes on the intradomain performance of AmFall/LW-CNN. (a) Moving variance window. (b) Moving sum window.

5) Impact of Window Sizes: AmFall uses the moving variance window for measuring the signal sensitivity to human activity and the moving sum window for specifying the period of human activity, of which the sizes are controlled using hyperparameters W in (4) and W_{m} in (13), respectively.[6]

Let us examine the effect that the sizes of these two windows have on the performance of AmFall/LW-CNN. Fig. 11(a) depicts the classification accuracy according to the moving variance window size, when the moving sum window size is fixed to 1.0 s. It is shown from Fig. 11(a) that the optimal size of the moving variance window depends on the dataset. This is because each dataset is usually collected under different noise conditions. A smaller W may be better for low impulse noise environments, where signal variation over short intervals is mainly originated from human activity. In Fig. 11(b), we explore the effect of the moving sum window size after fixing the moving variance window size to its best value from Fig. 11(a), i.e., 0.1 for A/A and 0.4 for D/D. It is observed from Fig. 11(b) that the moving sum window size has less impact on performance than the moving variance window size. Based on the observation from Fig. 11, we have set the moving variance window size to 0.1 s for the datasets A, B, and C, and to 0.4 s for the dataset D, and set the moving sum window size to 1.0 s for all datasets.

[5] When similar activities are regarded as the same activity, the unseen activity in C/AB and D/AB is *lying down*, whereas the unseen activities in AB/C are *jumping*, *random events*, *bending and picking up*, and *no event*. For AB/D, the *falling from a sitting position* and *picking up a pen* are unseen.

[6] The moving variance window size is $(2W+1)T_{\mathrm{s}}$ and the moving sum window size is $(2W_{\mathrm{m}}+1)T_{\mathrm{s}}$. Since the sampling rates of the datasets A and D are 1000 Hz and 320 Hz, respectively, $T_{\mathrm{s}} = 1.0$ ms for the dataset A and $T_{\mathrm{s}} = 3.125$ ms for the dataset D.

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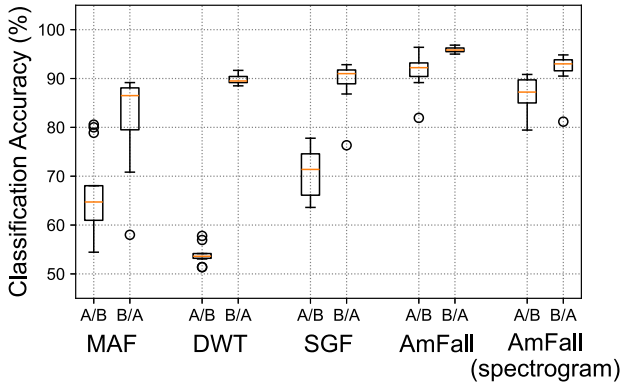


Fig. 12. Cross-domain classification performance of filtering-based scalogram images, AmFall preprocessed scalogram images, and AmFall preprocessed spectrogram images.

TABLE VI
COMPUTATIONAL TIME OF EACH STEP

Process	Time (ms)
CSI stream selection	14.6
PCA and PC selection	1.4
Scalogram calculation	39.3
Scalogram denoising	89.0
Scalogram speed estimation	16.5
Scalogram segmentation	2.3

TABLE VII
COMPLEXITY OF AMFALL-CNN MODELS

Model	Training time (ms)	Inference time (ms)	Number of parameters
AmFall/LW-CNN	43.2	1.7	56,322
AmFall/ResNet-18	155.7	5.9	11,177,538
AmFall/ResNet-34	243.0	8.4	21,285,698

C. Competitive Comparisons

1) Comparison With Filterings: It is noted that the proposed preprocessing technique achieves a denoising effect by selecting the PCs of high-quality CSI streams. Let us compare the denoising effects of our selection method and three filtering techniques widely used for denoising: moving average filter (MAF) [40], DWT [34], and Savitzky-Golay filter (SGF) [41]. The scalogram after filtering is represented as $\mathcal{S}_0(\sum_{i=1}^{N_H} \text{Filter}(|\mathbf{h}_i|))$, where Filter() is either MAF, DWT, or Savitzky–Golay filter (SGF). The parameters for each filtering method are set to achieve a fundamental stopband frequency of approximately 30–32 Hz,[7] given a sample frequency of 1000 Hz (MAF: window length of 32, DWT: 5-level sym4 wavelet, SGF: a 6th-degree polynomial with a 0.1 s span). The LW-CNN is used for classification. The dataset augmentation, scalogram denoising, and segmentation are not applied.

Fig. 12 shows the classification accuracy for the cross-domain scenarios, A/B and B/A. All of three filtering methods achieve higher accuracy in B/A compared to A/B, since the dataset B contains more noise than the dataset A. On the other hand, the filtering methods provide lower performance compared to AmFall's CSI preprocessing. This is because they remove not only noise but also useful information that may exist outside the stopband frequency range, whereas our preprocessing technique selects CSI streams and PCs to retain valuable information across a broader range of effective frequencies.

2) Scalogram Versus Spectrogram: A spectrogram also can be used for classification [20], [27]. To justify our use of CWT-based scalogram, we assess the classification performance using scalogram and spectrogram images for the same CSI stream $\mathbf{s}$ in (11). For a fair comparison, AmFall-specific methods such as the dataset augmentation, scalogram denoising, and segmentation are not applied. For the spectrogram image, the data is processed using the STFT with a window size of 512 ms and a window shift of 128 ms. The spectrogram image is generated ignoring frequency responses above 170 Hz, similar to the scalogram.

[7]Since most of the signal energy for human activities (including falls) is typically concentrated between 1 Hz and 30 Hz [27], we select this stopband frequency range.

The results of the two techniques using the LW-CNN classifier are presented on the right side of Fig. 12. The scalogram-based AmFall outperforms the spectrogram-based AmFall by 4.59% and 3.77% on average, for A/B and B/A. This confirms that the CWT is more suitable for fall detection than the STFT, at least for our datasets.

D. Computational Time and Parameter Size of CNN Models

Table VI presents the average computational time for each step of the proposed method, using the hardware and software specified in Section IV-A without GPU processors. The time of each step is the average time measured while generating a single scalogram image from each CSI observation with dimensions of $(8000 \times 2 \times 3 \times 30)$ for the 1x dataset. A significant portion of the total computational time comes from scalogram processing. However, the total preprocessing time is about 163 ms, which may be acceptable even with limited processing capabilities.

Table VII compares the complexity of the CNN models used by AmFall as a classifier. It lists the processing time per scalogram image using the Colab environment with the T4 GPU option. The LW-CNN has the $56,322$ parameters which are about $1/200$ and $1/400$ of those in ResNet-18 and ResNet-34, and it also requires approximately $1/3$ of ResNet-18 or $1/5$ of ResNet-34 in the processing time. Note that AmFall/LW-CNN provides comparable performance to AmFall/ResNet-18 and AmFall/ResNet-34. Because the AmFall preprocessing method can be flexibly coupled with most well known CNN models, any appropriate CNN model could be adopted, depending on available processing capabilities of deployment environment.

E. Comparison With Other Fall Detection Methods

AmFall support two types of classifiers, DL-based classifier and speed estimate-based classifier (see Fig. 1). Let us compare the performance of AmFall with those of the existing schemes.

1) DL-Based Classification: AmFall/LW-CNN is compared to three recent studies with high fall detection accuracy: Nakamura et al. [20], Chu et al. [21], and DapFall [22]. The

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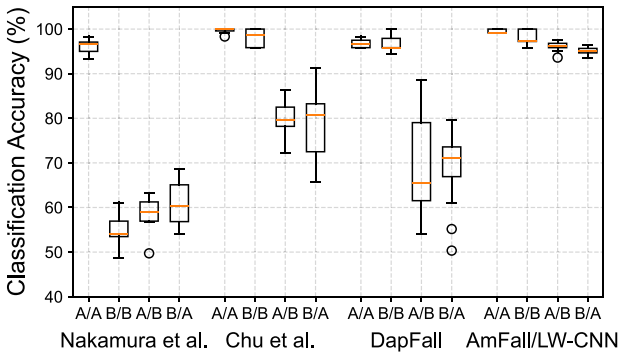


Fig. 13. Performance comparison of the proposed AmFall/LW-CNN with the methods of Nakamura et al. [20], Chu et al. [21], and DapFall [22].

augmentation for AmFall/LW-CNN is excluded to ensure fair comparison. Segmentation is also omitted for all methods.

Nakamura et al. applied PCA and select PCs in descending order of their eigenvalues until the cumulative ratio of eigenvalues exceeds a threshold. The spectrograms generated from the selected PCs using the STFT is combined into a single spectrogram by summing the individual spectrograms, weighted by their respective eigenvalue ratios. The resulting spectrogram is then normalized by the maximum value among all spectrograms within a given environment. ResNet-34 is used for classification.

Chu et al. optionally applied DWT-based denoising and perform downsampling by a factor of 8. A single image is generated from three Rx-antenna inputs, which are considered as RGB channels. After normalization, the image is input to EfficientNet-B0 [42] with specific customizations.

In DapFall, the amplitude matrix, consisting of subcarriers and time indices, is segmented into multiple submatrices using a sliding window. For each submatrix, the probability density function (PDF) of each subcarrier vector is estimated using Gaussian kernel density estimation, resulting in a PDF matrix. This PDF matrix is then converted into an RGB image, and the sequence of such images, corresponding to the submatrices, is used as a video-like input to the 3-D CNN model, R2+1D [43], for fall detection.

Fig. 13 shows the comparison results. The scheme of Nakamura et al. has very poor performance except in A/A. This may be primarily due to its PC selection process, which chooses the first PC and other major PCs sequentially until reaching a threshold. In cases that a dataset contains less noise, such as in the dataset A, some noise can be effectively mitigated by selecting major PCs because the first PC with the largest eigenvalue typically contains most of the useful information. However, when the noise level is relatively high like in domain B, a significant amount of noise may remain in the PCs with large eigenvalues. Specifically, the first PC is likely to contain the most noise and, in such cases, removing the first PC may be more beneficial. As a result, the scheme of Nakamura et al. that always selects the first PC provides poor performance for the dataset B.

The scheme of Chu et al. provides better performance than that of Nakamura et al., but it exhibits very poor performance in cross-domain scenarios. The scheme preserves features below 62.5 Hz by using 1/8 downsampling with a 1000 Hz sampling frequency. However, considering the theoretical maximum frequency of about 170 Hz for fall detection, 62.5 Hz may not be sufficient to detect falls in certain WiFi deployments. Furthermore, some CSI streams may contain significant noise but this scheme does not effectively eliminate noisy CSI streams. Accordingly, the scheme of Chu et al. suffers from performance degradation in noisy environment.

DapFall shows comparable performance in intradomain scenarios, but exhibits significantly lower performance in cross-domain evaluations. The design of DapFall may be ineffective for extracting domain-independent features in challenging NLOS environments.

Even with a light CNN classifier, the proposed scheme (i.e., AmFall/LW-CNN) achieves significantly higher performance than the comparison methods which use more complex CNN classifiers, i.e., EfficientNet-B0 with customizations for Chu et al., ResNet-34 for Nakamura et al., and R2+1D for DapFall.

TABLE VIII
SPEED ESTIMATE-BASED CLASSIFICATION (%)

Augmentation	Domain	Accuracy	Precision	Recall	F1 score
AmFall (1x)	A	95.33	93.31	97.67	95.44
	B	91.39	94.08	88.33	91.12
AmFall (6x)	A	97.83	97.05	98.67	97.85
	B	94.44	95.98	92.78	94.35
WiSpeed [19]	A	77.33	77.15	77.67	77.41
	B	59.17	59.43	57.78	58.59

2) Speed Estimate-Based Classification: The speed estimated from (14) reflects the change in propagation length caused by human torso movement. Therefore, its maximum value can be tested against a specific speed threshold to determine whether a fall has occurred.

Table VIII presents the speed estimate-based fall detection performance of AmFall. The speed estimates are extracted from the 1x and 6x datasets. Each speed estimate for the 6x dataset is the average of speed estimates of the six augmented versions derived from the 1x dataset. Since each domain has different propagation lengths, the threshold is applied on a per-domain basis. Table VIII presents an accuracy of more than 91.39%, regardless of the domains or the augmentation methods. The results using the 6x dataset are better than those of the 1x dataset, since the 6x dataset provides a more stable speed estimate by averaging the derived versions. Better performance is obtained in the domain *A* than in the domain *B*, due to lower noise levels in the domain A.

For performance comparison, we also present the performance of WiSpeed [19] in Table VIII. WiSpeed estimates speed by applying statistical theory of electromagnetic waves to CSI power and detects falls by thresholding the maximum estimated speed. In WiSpeed, the absence of denoising techniques may make speed estimates highly susceptible to noise, resulting in severe performance degradation in the noisy environment (i.e., B), as shown in Table VIII.

The classification based on speed estimate does not require a DL classifier. Moreover, the speed information itself is domain-independent. Therefore, this speed estimate-based fall

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detection can be useful in environments with limited processing capabilities or when prompt decision-making is required.

V. Conclusion

In this article, we have proposed a WiFi CSI amplitude-based fall detection system using denoised scalogram images, named AmFall. To address TTW conditions, we have developed a preprocessing method that adaptively selects CSI streams and PCs containing valuable information. We have also proposed a novel scalogram denoising technique that removes fall-irrelevant components and residual noise while retaining fall-related features. This denoising approach enables the extraction of speed estimates caused by human activities, which are subsequently used for scalogram segmentation and non-DL fall detection. To ensure sufficient training data, we have introduced a simple yet effective dataset augmentation to generate multiple scalograms from a single CSI observation. AmFall offers multiple options for a classifier, depending on the processing capabilities and limitations of the actual deployment environment: a non-DL approach using speed estimates, a lightweight CNN model for low-computing environments, and deeper CNN models for high-computing scenarios. Through the combined effect of these CSI preprocessing, scalogram denoising, and speed-based segmentation, AmFall significantly outperforms state-of-the-art WiFi-based fall detection methods in TTW conditions and cross-domain scenarios, demonstrating its usefulness as a practical fall detection system in real-world environments.

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